

- 32.a.i. Explain the Physical Vapour Deposition (PVD) method in detail. (8 Marks)
- ii. Write down the comparison between SEM and TEM. (4 Marks)

(OR)

- b. Write the principle of Atomic Force Microscopy (AFM). Explain basic components and working of AFM. Write the merits and demerits and applications of AFM.

Reg. No.

B.Tech. DEGREE EXAMINATION, MAY 2019
First / Second Semester

18PYB103J – PHYSICS : SEMICONDUCTOR PHYSICS

(For the candidates admitted during the academic year 2018-2019 onwards)

Note:

- (i) Part - A should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
- (ii) Part - B and Part - C should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)

Answer ALL Questions

- The free electrons in metals always obey
(A) Fermi-Dirac statistics (B) Widemann Franz law
(C) Bose Einstein statistics (D) Drude Lorentz theory
- At low temperatures, semiconductors will become
(A) Conductors (B) Insulators
(C) Ferroelectrics (D) superconductors
- The quantum of energy in elastic wave is known as
(A) Photon (B) Electron
(C) Phonon (D) Magnon
- The acceptor atoms in extrinsic p-type semiconductor contribute
(A) Holes to conduction band (B) Holes to valence band
(C) Electrons to conduction band (D) Electrons to valence band
- In a basic OLED structure, the diamine layer is used as a
(A) CCL (B) ETL
(C) ITL (D) HTL
- Electron-Hole pairs are produced due to _____
(A) Recombination (B) Thermal energy
(C) Ionization (D) Doping
- LED is a semiconductor p-n junction diode which converts _____ under forward bias
(A) Light energy into electrical energy (B) Thermal energy into electrical energy
(C) Electrical energy into light energy (D) Electrical energy into thermal energy
- A semiconductor has _____ temperature coefficient of resistance.
(A) Positive (B) Zero
(C) Large (D) Negative

9. The average velocity acquired by the electron in a particular direction during the presence of electric field is called _____.
 (A) Relaxation time (B) Drift velocity
 (C) Collision time (D) Diffusion current
10. The _____ does not require any external energy for process to take place.
 (A) Stimulated emission (B) Spontaneous emission
 (C) Stimulated absorption (D) Stimulated radiation
11. The Ratio of the maximum power that can be extracted from solar cell to the ideal power is called _____.
 (A) Short circuit voltage (B) Open circuit voltage
 (C) Fill factor (D) Drift voltage
12. The current density is directly proportional to the _____.
 (A) Rest mass of electron (B) Applied electric field
 (C) Density of core electron (D) Collision time
13. An ideal monochromator should have an _____ narrow effective band width
 (A) Infinitely (B) Small
 (C) Zero (D) Finite
14. In a _____, the electrons are ejected from a photoemissive surface and are amplified within the cell.
 (A) Photomultiplier tube (B) Bolometer
 (C) Electrode (D) Photodiode
15. Alkali metals and their oxides are best _____ materials.
 (A) Photoemissive (B) Conducting
 (C) Insulating (D) Semiconducting
16. _____ is used for separating source radiation wavelengths.
 (A) Antenna (B) Detector
 (C) Monochromator (D) Display device
17. The AFM tip should have a radius of curvature of _____.
 (A) Greater than 20 – 50 nm (B) Around 100 nm
 (C) More than 100 nm (D) Lesser than 20 – 50 nm
18. In a quantum wire, the material size is reduced
 (A) In three directions (B) In one direction
 (C) In two direction (D) In zero directions
19. Carbon nanotube reactivity is related to _____.
 (A) Volume (B) Length
 (C) Diameter (D) Width
20. In CVD chamber, the precursors are introduced to the reaction chamber in the _____ state.
 (A) Liquid (B) Solid
 (C) Gaseous (D) Semisolid

PART – B (5 × 4 = 20 Marks)
 Answer ANY FIVE Questions

21. What are the merits and demerits of quantum free electron theory?
22. Write down the Fermi distribution function. How does the function vary with temperature?
23. Write notes on photocurrent in p-n junction.
24. Explain the concept of drift current.
25. Describe about the optical recombination process.
26. Write a short note on photoemissive cell.
27. Write any four applications of carbon nanotube.

PART – C (5 × 12 = 60 Marks)
 Answer ALL Questions

- 28.a.i. What is Density of states? Derive an expression for density of states for a semiconducting material. (10 Marks)
- ii. Calculate the probability than an energy state above E_F is occupied by an electron. (2 Marks)
- (OR)
- b. Describe the behaviour of electrons in a periodic potential and hence explain the Kronig-Penny model.
29. a. Explain the principle, construction, working of OLED. Mention merits, demerits and applications.
- (OR)
- b. What is Intrinsic semiconductor? Explain atomic structure and energy level diagram of Intrinsic semiconductor. Where does the Fermi level lie in an Intrinsic semiconductor?
30. a. Explain the basic principle and applications of photovoltaic cell with their efficiency.
- (OR)
- b.i. Write about the basic assumptions of Drude and Lorentz model with their merits and demerits.
31. a. Write the principle and working of Deep Level Transient Spectroscopy (DLTS). Explain the basic components of DLTS.
- (OR)
- b. Explain the resistivity measurement by four probe methods by Linear method and Van der Paw method.